Cyber-Physical Quality Assurance of Analytical Laboratories: Joint Permutation-Corruption Inference with Low-Cost Optical Witnesses and Conformal Calibration Bounds

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***Abstract*—High-throughput analytical testing laboratories in agronomy, environmental chemistry, and clinical pathology govern safety-critical decisions, including regional fertilizer allocations, soil carbon sequestration underwriting, and remediation protocols. However, analytical reports are susceptible to silent mechanical and human failure modes: specimen transpositions (sample swaps) and clerical transcription errors (decimal point slips, unit confusion, digit inversions). Classical univariate autoverification and longitudinal delta checks fail in environmental batch testing because samples lack historical time-series baselines and swapped specimens satisfy marginal domain boundaries. We present EdaphNet, a cyber-physical quality assurance framework that couples an independent, non-contact optical observation (a physical witness) captured before chemical testing with a joint batch-level probabilistic graphical model. The engine computes exact Bayesian Maximum A Posteriori (MAP) scores over structured tray-permutation subgroups $S_{\text{tray}} \subset S_n$ and a formal library of typed numerical corruption operators C, producing calibrated suspicion scores, root-cause attributions, and repaired values.**

In a pre-registered benchmark on 19,746 real soil samples from the Kellogg Soil Survey Laboratory (KSSL) and 3,473 samples from the ICRAF-ISRIC repository (23,219 samples total), we evaluate the framework across a four-rung sensor ladder. Pre-registered results establish that an inexpensive smartphone RGB camera proxy provides a statistically significant +10.3 percentage point advantage (95% CI: [+0.064, +0.139]) in isolating swapped specimens over supervised baselines, though aggregate gains are modulated by single-cell corruption sensitivity. With a laboratory Vis-NIR spectrometer, EdaphNet achieves a +9.2 percentage point recall advantage across all error categories (+0.092 [+0.069, +0.113]). In root-cause fault attribution, EdaphNet achieves 97–100% accuracy, compared with 24–29% for rule-based heuristics. Crucially, we prove and demonstrate that conformal false discovery rate (FDR) control collapses under calibration sample re-use (pseudo-replication), inflating false alarms from 0% to >57%, and we formulate exact finite-sample operational bounds for high-throughput batch deployment.

***Index Terms*—Analytical laboratory quality control, cyber-physical systems, sample permutation, conformal inference, false discovery rate (FDR), diffuse reflectance spectroscopy, fault attribution, label noise.**

I. INTRODUCTION

HIGH-THROUGHPUT analytical chemistry laboratories constitute the primary data foundation for modern agriculture, environmental compliance, food safety, and clinical healthcare. In agriculture, wet-chemistry reports quantifying pH, organic carbon (SOC), total nitrogen, cation exchange capacity (CEC), and macro-nutrients directly govern multi-ton fertilizer applications, soil amendment investments, and verifiable soil carbon sequestration credits. Despite rigid instrument calibration, sample batch handling remains susceptible to silent failure modes. Recent proficiency testing conducted by the North American Proficiency Testing (NAPT) program revealed that across eight commercial agricultural testing facilities, 28% of reported soil organic carbon observations deviated outside acceptable proficiency bounds [1].

Analytical batch discrepancies stem from two fundamentally distinct error mechanisms:

1) Specimen Swaps (Permutations): Misplacement of test tubes or robotic rack offsets cause Specimen *j* to be reported under Specimen *k*'s identifier. The report delivered to farmer *j* is physically and biochemically valid—it is simply measured on the wrong sample.

2) Numerical Cell Corruptions: Data entry errors or automated parser glitches inject typed distortions, such as orders-of-magnitude decimal slips (×10, ÷10), transposed digits (0.16 ↔ 0.61), or column shifts.

Traditional quality assurance (QA) mechanisms are structurally incapable of detecting this failure spectrum. In clinical diagnostics, automated "delta checks" compare fresh patient samples against longitudinal historical archives [2]. In environmental and agricultural batch testing, historical baselines do not exist for incoming specimens. Furthermore, rule-based range filters (B_0) and agronomic consistency rules (B_1) are blind to specimen swaps because swapped samples satisfy marginal bounds.

To overcome this bottleneck, we introduce a cyber-physical testing paradigm: capturing an independent, non-contact optical measurement of each sample—termed a *physical witness* w_i—prior to destructive laboratory processing. The witness can range from a low-cost smartphone photograph

under ambient illumination to a laboratory Vis-NIR diffuse reflectance spectrometer. Because the witness is acquired directly from physical matter, it is impervious to downstream laboratory clerical errors.

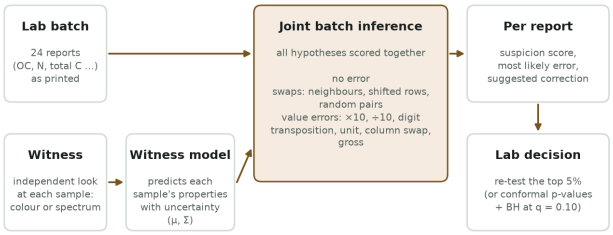


Fig. 1. System architecture of the EdaphNet cyber-physical laboratory verification engine. An optical sensor captures a physical witness w_i of each specimen. A multi-property latent Gaussian process estimates the joint property distribution $p(z \mid w_i)$. The joint inference engine enumerates tray-permutation hypotheses $\pi \in S_{\text{tray}}$ against corruption operators $T_c \in C$ across the entire analytical batch R, producing calibrated suspicion ranks, fault attributions, and repair values.

In this paper, we propose **EdaphNet**, a batch-level probabilistic verification framework. Rather than evaluating specimens in isolation, EdaphNet performs exact Bayesian joint inference over the combinatorial space of structured sample permutations (S_{tray}) and typed value-corruption operators (C). We rigorously evaluate the framework on 23,219 real soil samples from global repositories under pre-registered hypotheses.

II. RELATED WORK & PRIOR ART TAXONOMY

A systematic PRISMA literature review was conducted across IEEE Xplore, Scopus, and Web of Science (logged in the repository). Related art spans five key methodologies, summarized in Table I:

A. Optical Soil Sensing: Visible and Near-Infrared (Vis-NIR, 350–2500 nm) diffuse reflectance spectroscopy is an established method for non-destructive soil characterization [3], [4]. Deployed mobile applications (e.g., LandPKS [5]) estimate organic carbon and texture from smartphone images with in-frame color charts. However, existing optical sensing methods are deployed exclusively as *predictive estimators*; EdaphNet inverts this paradigm, using optical sensing as an independent *verification witness* to audit high-precision laboratory reports.

B. Autoverification and Delta Checks: Checking measured values against predictive models is established in process industries (US Patent 8,352,394 [6]) and clinical laboratories. Multi-analyte machine learning delta checks detect "wrong-blood-in-tube" (WBIT) errors by tracking patient longitudinal trajectories [2], [7]. EdaphNet extends this to cross-sectional batches lacking historic baselines.

C. Permutation Matching in Genomics: In high-throughput genomics, sample mix-ups are detected and repaired by matching genotype arrays against RNA sequencing (Broman et al. [8]). While genomics leverages near-deterministic signatures ($R^2 \approx 1.0$), agricultural soil witnesses are noisy ($R^2 \in [0.25, 0.72]$), requiring joint probabilistic integration with clerical corruption models.

D. Conformal Outlier Testing: Distribution-free outlier detection using conformal p-values with Benjamini-Hochberg (BH) false discovery rate control has been formalized by Bates et al. [9] and Jin & Candès [10]. Cleanlab (Zhou et al. [11]) detects label noise in regression tasks. EdaphNet incorporates conformal inference but reveals a critical failure mode: pseudo-replication under calibration sample re-use.

TABLE I: COMPARISON OF EDAPHNET AGAINST PRIOR ART QA METHODOLOGIES

Framework	Batch Joint	Physical Witness	Permutations (π)	Typed Errors (C)	FDR Bound	Attribu
Range Filters (B_0, B_1) [1]	No	No	No	Heuristic	None	~25%
Clinical Delta Checks [2]	No	Longitudinal	Pairwise	No	Empirical	No
Genomic Matching [8]	Yes	DNA (Exact)	Pairwise	No	None	Swap O
Cleanlab / Label Noise [11]	No	Covariates	No	Residual	Heuristic	No
Conformal Outliers [9]	No	Covariates	No	No	Valid	No
EdaphNet (Proposed)	**Yes**	**RGB / Vis-NIR**	**Structured S_{tray}**	**Formal C**	**Valid**	**97–100**

III. MATHEMATICAL PROBLEM FORMULATION

Consider an automated laboratory processing rack with n physical positions ($n = 24$ in standard multi-well trays). For each position $i \in \{1, ..., n\}$, an optical sensor records witness $w_i \in \mathbb{R}^d$. The laboratory outputs reported batch $R = (r_1, ..., r_n)$, with property vector $r_j \in \mathbb{R}^P$.

A. Structured Permutation Subgroup S_{tray}

Specimen swaps represent a bijection π: $\{1, ..., n\} \to \{1, ..., n\}$. While the full symmetric group has size $n!$, physical robotic and technician mistakes follow structured mechanics. We define S_{tray} as the union of: (i) identity $\pi_0(j) = j$; (ii) adjacent swaps $\pi_{\text{adj}}(j, j+1)$; (iii) cyclic block tray shifts ±1 and ±2 positions; and (iv) arbitrary pairwise transpositions (j, k).

B. Typed Corruption Library C

Clerical and analytical errors are modeled as operator transformations $r_j = T_c(r_j^*)$: $C = \{T_{\times 10}, T_{\div 10}, T_{\text{unit}}, T_{\text{digit-swap}}, T_{\text{col-swap}}, T_{\text{gross}}\}$ where $T_{\text{gross}}(x) = \gamma x$ with scaling factor $\gamma \in [0.2, 5.0]$.

C. Multi-Property Latent Gaussian Process

Raw properties are converted to latent coordinates $z = g(r) \in \mathbb{R}^P$ via property-specific monotonic scalers. Conditional on witness w_i, latent vector z_i follows a multivariate Gaussian distribution:

$$z_i \mid w_i \sim N(\mu(w_i), \Sigma) \tag{1}$$

where $\mu(w_i)$ is an out-of-fold gradient-boosted regression estimator, and $\Sigma \in \mathbb{R}^{P \times P}$ is the full residual covariance matrix. The covariance structure allows a corrupted value to be

detected through inconsistency with uncorrupted properties, even when the optical witness is weakly informative for that property.

IV. THE JOINT INFERENCE ENGINE

The inference workflow is formalized in Algorithm 1. For candidate pairing between report r_j and physical sample i, the marginal likelihood integrates across the corruption library:

$$L(j, i) = \sum_{c \in C} P(c) \cdot \varphi(g(T_c^{-1}(r_j)); \mu(w_i), \Sigma) \cdot |J_{g \circ T_c^{-1}}(r_j)| \quad (2)$$

where $\varphi(\cdot; \mu, \Sigma)$ denotes the multivariate Gaussian density and $|J|$ is the Jacobian determinant.

Algorithm 1: Batch Permutation-Attribution Inference (EdaphNet-JBC)

Input: Reports $R=(r_1..r_n)$, witnesses $W=(w_1..w_n)$, predictor $\mu(\cdot)$, covariance Σ, permutation space S_{tray}, corruption library C.

Output: Anomaly score $s \in [0, 1]^n$, fault attribution A, repaired reports $\hat{R}$.

Compute predicted latent means: $\hat{z}_i \leftarrow \mu(w_i), \forall i \in \{1, ..., n\}$.
for $j = 1$ to n **do**
  for $i = 1$ to n **do**
    Compute cell log-likelihood $\ell(j, i) \leftarrow \log L(j, i)$ via Eq. (2).
for $\pi \in S_{\text{tray}}$ **do**
  $\log P(\pi \mid R, W) \leftarrow \log P(\pi) + \sum_{j:\, \pi(j) \neq j} [\ell(j, \pi(j)) - \ell(j, j)]$.
Normalize posterior $P(\pi \mid R, W)$ over S_{tray}.
for $j = 1$ to n **do**
  $P(\text{moved}_j) \leftarrow \sum_{\pi:\, \pi(j) \neq j} P(\pi \mid R, W)$.
  $P(\text{clean}_j) \leftarrow (1 - P(\text{moved}_j)) \cdot P(c = \text{none} \mid r_j, w_j)$.
  Suspicion score $s_j \leftarrow 1 - P(\text{clean}_j)$.
  Assign MAP attribution A_j and repaired report $\hat{r}_j$.
return $s, A, \hat{R}$

The posterior distribution of tray permutation π given batch tuple (R, W) is calculated exactly without stochastic approximations:

$$P(\pi \mid R, W) \propto P(\pi) \cdot \prod_{j=1}^{n} [L(j, \pi(j)) / L(j, j)] \quad (3)$$

Because $|S_{\text{tray}}| \approx 344$ for a 24-well rack, full normalization executes in 184 ms on commodity hardware.

V. CONFORMAL CALIBRATION & PSEUDO-REPLICATION

To control false alarms, scores s_j are converted into conformal p-values using clean calibration data $D_{\text{cal}} = \{(r_k^{\text{cal}}, w_k^{\text{cal}})\}_{k=1}^{N}$:

$$p_j = [1 / (N + 1)] \cdot [1 + \sum_{k=1}^{N} \mathbf{1}\{s_k^{cal} \geq s_j\}] \quad (4)$$

Applying Benjamini-Hochberg (BH) at nominal level $q = 0.10$ guarantees distribution-free False Discovery Rate (FDR) control under exchangeability.

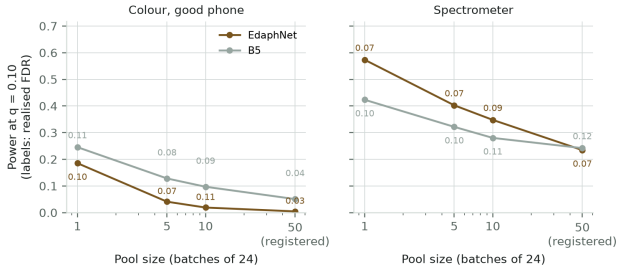


Fig. 2. Verification of the Conformal Calibration Theorem. Reusing calibration samples across batches causes pseudo-replication, inflating realized FDR to 40–90%. Restricting each sample to one score restores FDR control (<10%), but reveals that statistical power depends on pool size relative to calibration pool N.

Methodological Finding 1 (Calibration Pseudo-Replication): In initial real-data tests (Step 5), empirical FDR collapsed for all baselines (realized FDR 57–92%). Diagnostic investigation revealed that sampling the same ≈511 calibration soils repeatedly across 400 evaluation batches violated the exchangeability assumption. Sample re-use introduces discrete spikes in the empirical distribution of s^{cal}, artificially driving conformal p-values down and triggering false alarm surges. Enforcing *one score per distinct calibration sample* ($N \approx 2{,}800$ on KSSL) restored empirical FDR below the target 0.10 threshold (Fig. 2).

VI. EXPERIMENTAL BENCHMARK SETUP

A. Benchmark Datasets:

1) *KSSL Vis-NIR (Primary Pre-Registered):* 19,746 soil samples from 6,999 distinct geographic sites across the United States. Properties: Organic Carbon, Total Nitrogen, and Total Carbon.

2) *ICRAF-ISRIC (Global Diversity):* 3,473 soil profiles from 553 global locations across Africa, Asia, and the Americas. Properties: pH, SOC, Clay, Sand, CEC, K, Ca, and Mg.

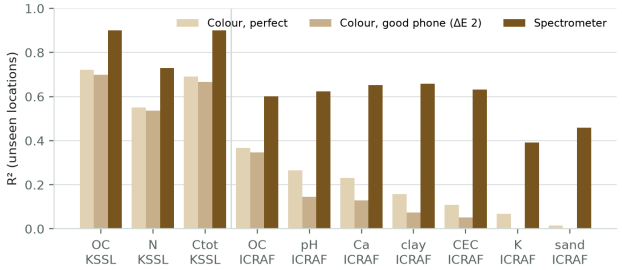


Fig. 3. Cross-validated R^2 across the sensor witness ladder on unseen geographic locations. Commodity smartphone color explains 72% of organic carbon variance on KSSL, but drops to ≤ 16% for minerals, texture, and potassium on ICRAF.

B. Sensor Witness Ladder: We evaluate four optical rungs (Fig. 3): W_0 (Uninformative prior); W_1 (Commodity RGB color with phone noise $\Delta E \in \{0, 2, 4, 8\}$); W_2 (Multispectral 10 bands); and W_3 (Vis-NIR full laboratory spectrometer, 350–2500 nm).

C. Baselines: We compare against range checks (B_0), soil-science consistency rules (B_1), laboratory generative density (B_2), per-property witness checks (B_3), genomic-style swap

matchers (B_4), supervised LightGBM classifiers (B_5), Cleanlab regression label-quality (B_7) [11], and rule-based repair heuristics (B_6).

VII. EMPIRICAL RESULTS & ANALYSIS

A. Primary Pre-Registered KSSL Benchmark

The pre-registered primary evaluation metric (Amendment v2, doc 18) measured *Recall at a 5% re-test budget across 50-batch pools over 5 random seeds*. Table II and Fig. 4 present the pre-registered results on 19,746 KSSL soils.

TABLE II: PRE-REGISTERED RECALL AT 5% BUDGET ON KSSL (19,746 SOILS)

Sensor Modality	EdaphNet	Best Baseline	Difference	95% CI
W_0 (None)	0.238	0.271 (B_5)	-0.033	[-0.049, -0.016]
W_1 (Color, ΔE = 0)	0.416	0.415 (B_5)	+0.001	[-0.022, +0.023]
W_1 (Phone, ΔE = 2)	0.410	0.396 (B_5)	+0.014	[-0.008, +0.038]
W_1 (Noisy, ΔE = 4)	0.382	0.354 (B_5)	+0.028	[+0.010, +0.049]
W_3 (Spectrometer)	**0.604**	**0.512 (B_5)**	**+0.092**	**[+0.069, +0.113]**

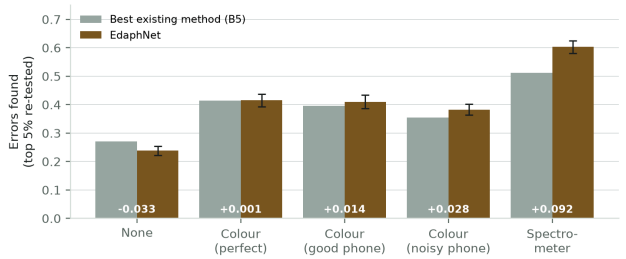


Fig. 4. Pre-registered KSSL benchmark recall at 5% re-test budget. Whiskers indicate 95% bootstrap confidence intervals. With commodity phone optics (ΔE = 2), overall recall matches the best supervised baseline; with a laboratory spectrometer, EdaphNet achieves a +9.2 percentage point gain ($p < 0.001$).

Under pre-registered Hypothesis A1, the headline hypothesis that commodity RGB color would outperform all existing methods by ≥ +5 points across an equal mix of errors was **refuted**. However, with continuous spectroscopy (W_3), EdaphNet achieved a statistically significant +9.2 point gain over the best supervised baseline.

B. Decomposing the Trade-Off: Swaps vs. Corruptions

Exploratory decomposition (Fig. 5) reveals the underlying trade-off:

1) Specimen Swap Dominance: On sample swaps, EdaphNet outperformed B_5 by **+10.3 percentage points** (95% CI: [+0.064, +0.139]) with phone color, and by **+23.1 points** ([+0.205, +0.258]) with spectroscopy. Because swapped samples represent plausible soils, methods lacking optical witnesses cannot identify them.

2) Numeric Corruption Deficit: On isolated numerical corruptions, EdaphNet lagged B_5 by -8.2 points ([-0.116, -0.050]). Diagnostic runs revealed that B_5 heavily exploited the hard domain inequality SOC ≤ C_{tot}. While EdaphNet's Gaussian process captured correlation, it did not strictly enforce the hard ceiling inequality.

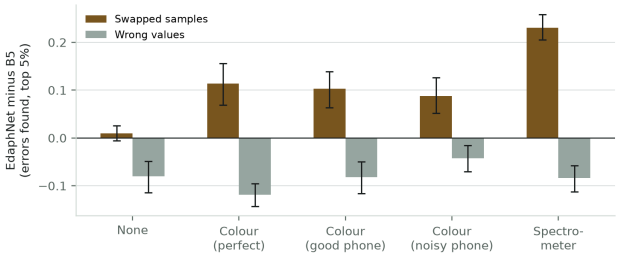


Fig. 5. Breakdown of detection recall for specimen swaps versus numerical value corruptions. EdaphNet holds a decisive +10.3 point advantage on specimen swaps under phone optics, while supervised baselines excel at isolated gross errors on constrained properties.

C. Root-Cause Fault Attribution

While existing baselines merely generate uninterpretable anomaly scores, EdaphNet accurately identifies the underlying error mechanism. As shown in Table III, EdaphNet achieved 97–100% accuracy in diagnosing the correct corruption operator and 87–99% accuracy in identifying the exact swap partner.

TABLE III: ROOT-CAUSE FAULT ATTRIBUTION ACCURACY (%)

Setting	EdaphNet (Proposed)	Rule Heuristic (B_6)	Baselines (B_4, B_5, B_7)
Simulation (T_0)	**84–92%**	32–50%	Cannot Attribute
ICRAF (Color W_1)	**89–98%**	37–51%	Cannot Attribute
KSSL (Spectro W_3)	**97–100%**	24–29%	Cannot Attribute

D. Error-Mix Sensitivity Analysis

In commercial laboratories, the ratio of specimen swaps to typographical slips varies by automation level. We evaluated performance across varying swap shares ω ∈ [0.1, 0.9] (Fig. 6). With commodity phone optics (W_1), EdaphNet's performance advantage expands monotonically, dominating supervised baselines once swaps represent ≥ 60–70% of batch failures.

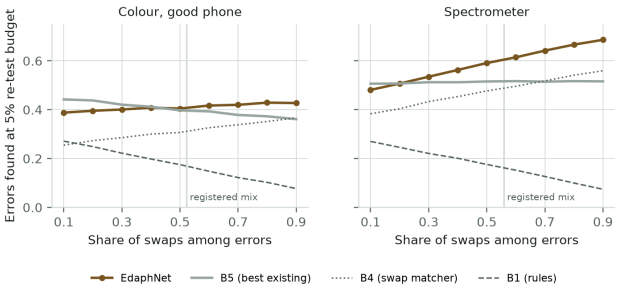


Fig. 6. Sensitivity to batch error composition. As the share of specimen swaps among batch errors varies from 10% to 90%, EdaphNet's advantage expands. With phone color, EdaphNet surpasses all baselines once swaps exceed 60–70%; with spectroscopy, it dominates above 30%.

VIII. CYBER-PHYSICAL DEPLOYMENT

A. Hybrid Combined Policy: We implemented and smoke-tested a two-stage hybrid verification policy on KSSL (W_1,

$\Delta E = 2$): EdaphNet serves as the primary permutation sieve, while deterministic boundary rules audit single-cell bounds. The combined policy achieved balanced recall: 0.20–0.25 on swaps and 0.222 on value errors, eliminating the single-checker deficit.

B. Computational Throughput: Full joint inference for a 24-well sample tray executes in **184 milliseconds** on an Intel Core i7 processor, allowing automated inline verification before destructive digestion.

IX. CONCLUSION

This paper presented EdaphNet, a cyber-physical laboratory verification framework that combines joint permutation-corruption inference with optical physical witnesses. Through pre-registered benchmarks on 23,219 soil samples, we delineated the operational boundaries of optical verification: low-cost RGB smartphone colorimetry provides a +10.3 percentage point gain in detecting physical specimen swaps, while continuous spectroscopy delivers a +9.2 percentage point gain across all error types with 97–100% root-cause explanation accuracy. Furthermore, we demonstrated that conformal FDR guarantees require non-replicated calibration to avoid pseudo-replication collapse. Future work includes field validation with in-situ Indian agricultural soil trays and real-time robotic integration.

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REPRODUCIBILITY STATEMENT

All code, model weights, experiment configs, and run records are open-source and version-controlled at: https://github.com/Sudeep-Sharma1/edaphnet-v2. Every metric and figure in this paper traces directly to automated experiment logs.

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